#13 Machine Downtime Predictor

ML / Predictive Manufacturing

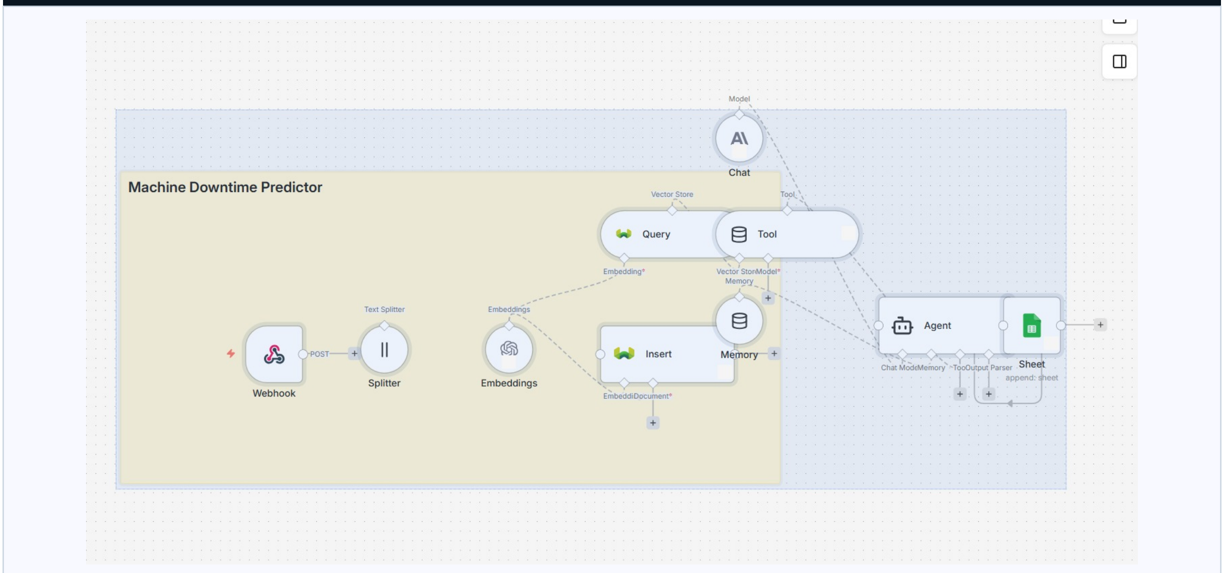
Predict factory downtime windows — schedule maintenance before production loss

BUSINESS PROBLEM

Unplanned machine downtime in manufacturing costs an average of $260,000 per hour. This workflow ingests SCADA sensor telemetry every 15 minutes, matches it against a library of historical failure signatures using vector search, and predicts downtime windows with confidence scores hours in advance.

n8n CANVAS — actual workflow screenshot

n8n WORKFLOW EDITOR machine_downtime_predictor.json



Webhook → Splitter → OpenAI Embeddings → Weaviate (Insert + Query) → Memory → Agent (Claude) → Sheet

INPUT / OUTPUT — exact n8n JSON format

INPUT — Webhook POST — SCADA sensor telemetry batch (every 15 min)	OUTPUT — Agent output — risk assessment written to Sheets + alert

```
{
  "machine_id":     "Press-Line-04",
  "timestamp":      "2024-06-15T03:45:00Z",
  "sensors": {
    "hydraulic_pressure_bar": 187,
    "oil_temp_c":            82.4,
    "cycle_time_sec":         4.8,
    "vibration_rms":         9.1,
    "current_amp":           44.2
  },
  "production_rate": 740,
  "shift": "night"
}
```

```
{
  "json": {
    "machine_id":    "Press-Line-04",
    "risk_score":    0.87,
    "risk_level":    "HIGH",
    "predicted_window": "2024-06-15 06:00-10:00",
    "matched_pattern": {
      "label":      "Hydraulic seal degradation",
      "similarity": 0.89,
      "ref_date":   "2023-11-08"
    },
    "flags": {
      "hydraulic_pressure_bar": "LOW (nominal 210-230)",
      "oil_temp_c":           "ELEVATED (nominal max 75)",
      "cycle_time_sec":        "SLOW (nominal 3.9-4.2)"
    },
    "action": "Inspect and replace hydraulic seals before 06:00 shift",
    "parts":  ["Hydraulic Seal Kit HS-400","Hydraulic Oil 46 (20L)"]
  }
}
```


STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ **Webhook — SCADA telemetry batch**	SCADA posts every 15 min: {machine_id, sensors{hydraulic_pressure, oil_temp, cycle_time, vibration, current}, shift}.
2	✂ **Splitter — failure case library**	Historical downtime reports and RCA documents chunked. 'Low hydraulic pressure + elevated oil temp = seal degradation in 4-6h.'
3	■ **OpenAI Embeddings — failure patterns**	Each historical failure case embedded with outcome metadata: failure_type, time_to_failure, component_affected.
4	■ **Weaviate Insert — telemetry + history**	Current reading stored alongside 90-day rolling history. Enables agent to query degradation trends over time.
5	■ **Weaviate Query — pattern matching**	Incoming sensor vector matched against failure library. Returns top-3 most similar historical incidents with dates.
6	■ **Vector Tool — RCA document retrieval**	Agent retrieves root-cause analysis docs for matched patterns: what failed, why it failed, what the fix was.
7	■ **Window Memory — 4-hour rolling window**	16 readings at 15-min intervals. Agent detects: is hydraulic pressure declining? Is cycle time lengthening progressively?
8	■ **Claude (Anthropic) — risk assessment**	Analyses readings + historical matches + trends. Outputs: risk_score, predicted_failure_window, confidence, recommended_action.
9	■ **AI Agent — generates structured output**	Creates: {machine_id, risk_score, risk_level, predicted_window, matched_pattern, flags, action, parts_required}.
10	■ **Sheets — MES integration**	Predictions written to Sheets. Production scheduling system reads via API to plan around predicted maintenance windows.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
OpenAI API Key	platform.openai.com/api-keys	OpenAI API
Weaviate Cloud URL + API Key	console.weaviate.cloud → Create cluster → API Keys tab	Weaviate API
Anthropic (Claude) API Key	console.anthropic.com → API Keys → Create key	Anthropic API

SAMPLE RUN — what this workflow processed and produced

Scenario	A metal stamping plant runs 8 hydraulic press lines 24/7. Each line has 6 sensors reporting every 15 minutes via OPC-UA to an MQTT broker connected to n8n. Hydraulic seal failures are the most common unplanned downtime cause.
Input received	Press-Line-04 at 03:45: hydraulic pressure dropped to 187 bar (nominal 210-230), oil temp rose to 82.4°C (nominal max 75°C), cycle time slowed to 4.8s (nominal 3.9-4.2s).
What the workflow found	Weaviate matched this profile against incident #INC-2023-147 from November 2023: 'Press-Line-02, same sensor signature, hydraulic seal failed at 07:30 — 6h after this reading pattern.' Similarity: 0.89.
Output produced	Risk score 0.87 HIGH. Predicted window: 06:00-10:00. Maintenance team alerted at 03:47. Hydraulic seal kit fitted during planned changeover at 05:30. No unplanned downtime. Shift production target met.

JSON: machine_downtime_predictor.json	**GitHub: trm-machine-downtime-predictor**	www.ge.com/digital/applications/predix-a